Robert Masen

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Work

Vultr — July 2025 - Current

Staff Software Engineer

- Ported a Virtual Machine Metadata server from PHP to Rust
- Lead an effort to transition a proof of concept rust hypervisor project to a full cargo workspace with multiple binary projects and large amounts of shared code
- Developed a kernel patch to provide network disk performance metrics via the sys-fs debug interface for accessing performance of a virtual machine guest from the host
- Designed and developed a system for linux virtual machine hosts to report guest resource allocation in real time
- Using real-time data published by virtual machine hosts for evaluating fleet wide capacity and selecting the most appropriate host for that customer, including pinned CPU and GPU offerings.

SmartThings — March 2020 - July 2025

Staff Software Engineer

- Implementing a memory efficient, Rust metrics collector for an embedded linux target
- Develop and maintain a Lua sandbox in Rust for executing untrusted code on embedded linux devices using gRPC for interactions with the wider platform
- Built an HTTP API in Rust running on edge devices to act as a cloud service
- To address performance issues on embedded linux hardware, developed and transitioned the platform to use a purpose build RPC mechanism to replace Tonic and gRPC.
- Develop a significant number of Rust platform utility and services to address shared needs across multiple teams
- Using a mixture of `rust-doc`, Confluence, PlantUML, Mermaid.js, and Lucid Chart I documented existing behavior, communicated proposed changes, and design documents for future opportunities.
- Mentored an Associate Software Engineer with the goal of getting promoted

Solution Design Group — February 2019 - March 2020

Software Development Consultant

- Build new features into a ASP.net payment portal
- Introduce static analysis utilities to increase code quality and consistency
- Started a Programming Languages Community

United Subcontractors — August 2017 - December 2018

Software Developer

- Build new features and perform bug fixes on an existing ASP.NET Web Forms application
- Transition existing application functionality into a React application using Typescript

Open Source

Rusty ECMA — JavaScript Parsing/Generating libraries

- RESAST: JavaScript AST in Rust
- RESSA: JavaScript Parser in Rust
- RESS: JavaScript Tokenizer in Rust
- RES-Regex: A parser of JavaScript Regular Expression Literals
- RESW: JavaScript Writer over RESAST data structures

Cosock — Lua couroutine I/O libraries

- Lustre: Websockets in Lua
- Luncheon: HTTP Parsing, Serialization and Types
- Luxure: HTTP Framework

Luminary — A compiler for Lua

- As an exercise in learning more about compilers, this project is a Lua front-end for LLVM.

Education

Eastern Michigan University

- Bachelor of Science in Communications

Minneapolis Community Education

- Associates in Software Development (Coursework)